

Maharashtra Public Service Commission Assessment

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Assistant Professor Electrical Engineering Group A

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Assistant Professor Electrical Engineering Group A

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15056110

Question Shuffling Allowed :

Yes

Question Number : 1 Question Id : 150561403 Question Type : MCQ Option Shuffling : No

The volume of the solid bounded by the surfaces : $z = 0$, $x^2 + y^2 = 1$ and $x + y + z = 3$ is _____.

Options :

1. 2π

2. 3π

3. $\frac{1}{3}\pi$

4. π

Question Number : 2 Question Id : 150561404 Question Type : MCQ Option Shuffling : No

The probability of getting 4 heads in 6 tosses of a fair coin is _____.

Options :

1. $\frac{15}{36}$

2. $\frac{1}{8}$

3. $\frac{8}{36}$

4. $\frac{15}{64}$

Question Number : 3 Question Id : 150561405 Question Type : MCQ Option Shuffling : No

Which of the following is the differential equation whose set of independent solutions is $[e^x, xe^x]$?

Options :

1. $y - 2y_1 + y_2 = 0$

2. $2y_1 + y + y_2 = 0$

3. $y + 2y_1 + y_2 = 0$

4. $y + 2y_1 - y_2 = 0$

Question Number : 4 Question Id : 150561406 Question Type : MCQ Option Shuffling : No

Which of the following is the solution of the homogeneous equation $(x + 2y) (dx - dy) = dx + dy$?

Options :

1. $2 \log (3x - 6y + 1)$
2. $2 \log (3x + 6y + 1)$
3. $2 \log (3x - 6y - 1)$
4. $2 \log (3x + 6y - 1)$

Question Number : 5 Question Id : 150561407 Question Type : MCQ Option Shuffling : No

The median of 6, 8, 9, 10, 11, 12, 13 is _____.

Options :

1. 6
2. 7
3. 10
4. 12

Question Number : 6 Question Id : 150561408 Question Type : MCQ Option Shuffling : No

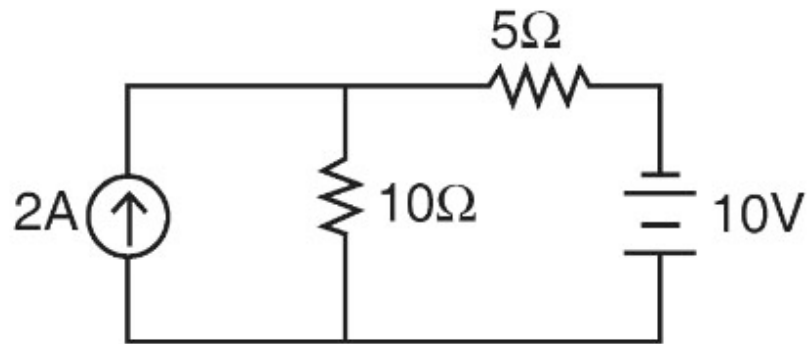
When three equal resistances are connected in delta, the equivalent star resistance per phase is given by _____.

Options :

1. $(\text{Resistance in delta})/3$
2. $(\text{Resistance in delta}) \times 3$
3. $\frac{2}{3} \times (\text{Resistance in delta})$
4. $\frac{3}{2} (\text{Resistance in delta})$

Question Number : 7 Question Id : 150561409 Question Type : MCQ Option Shuffling : No

Find the current in 10Ω resistor using super position theorem.

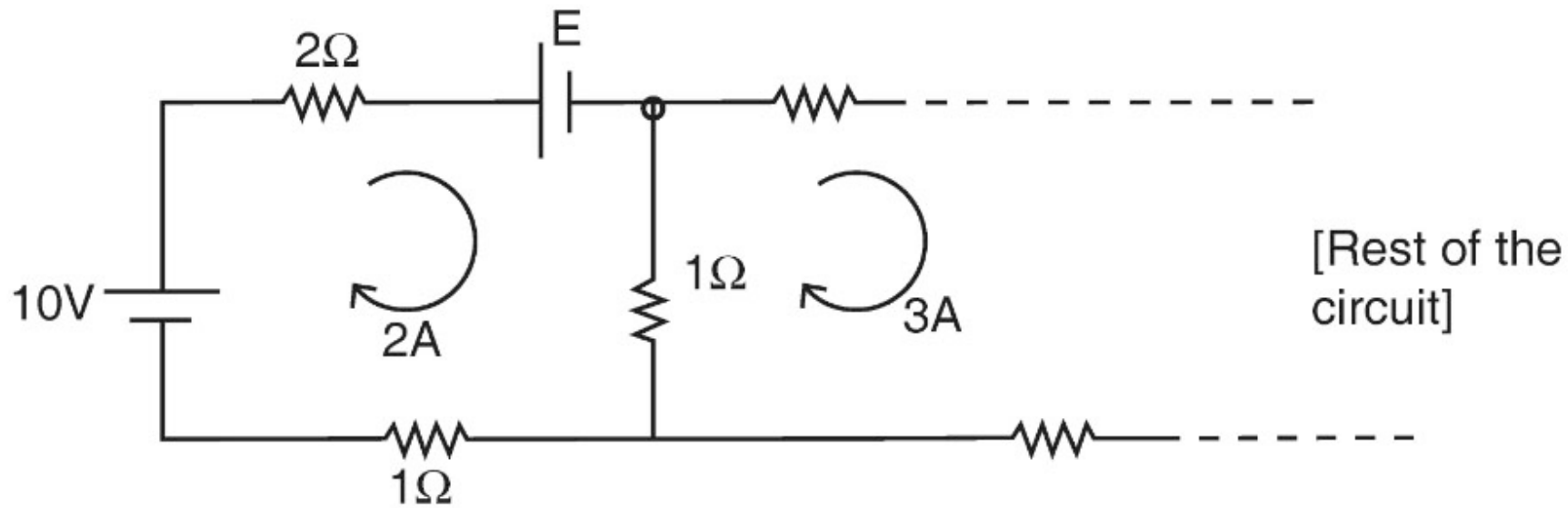


Options :

1. Zero
2. $\frac{4}{3}$ A downwards
3. $\frac{4}{3}$ A upwards
4. None of these

Question Number : 8 Question Id : 150561410 Question Type : MCQ Option Shuffling : No

Find unknown emf E in the following circuit using mesh analysis.

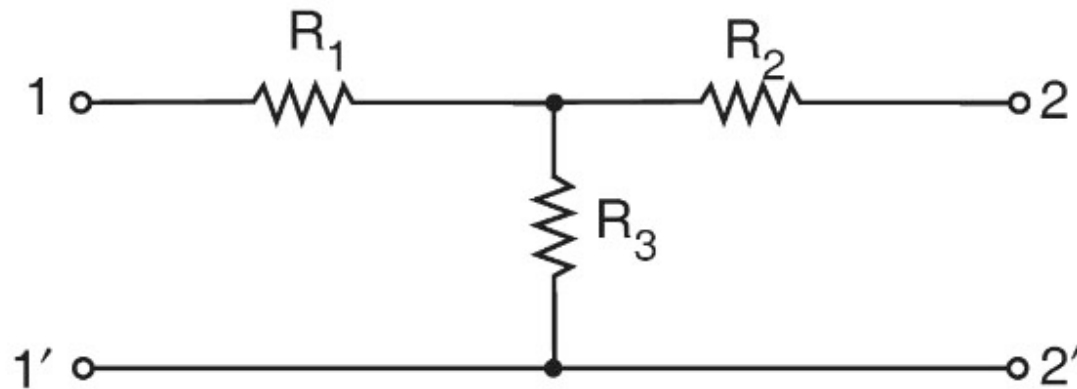


Options :

1. $3V$
2. $5V$
3. $4V$
4. $6V$

The open circuit impedance parameter (z-parameter) matrix for the network given

below is $Z = \begin{bmatrix} 10 & 5 \\ 5 & 15 \end{bmatrix}$



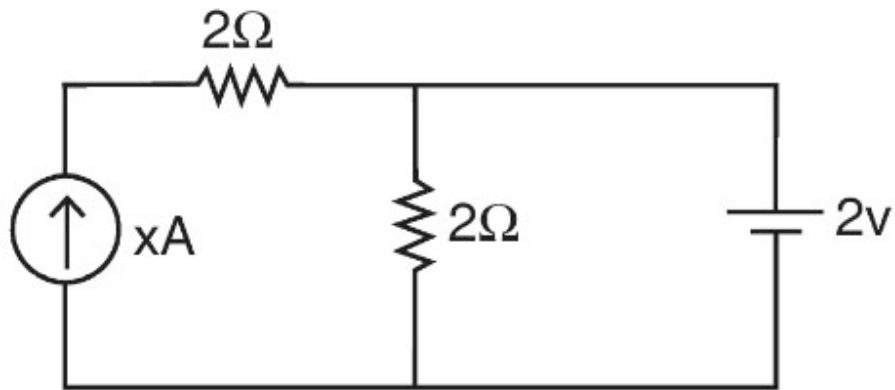
The value of resistance R_2 is _____.

Options :

1. 5Ω
2. 10Ω
3. 15Ω
4. 20Ω

Question Number : 10 Question Id : 150561412 Question Type : MCQ Option Shuffling : No

The total power consumed in the circuit shown below is 10 Watts.



The value of source current x will be _____.

Options :

1. 1A
2. 2A
3. 3A
4. 4A

Question Number : 11 Question Id : 150561413 Question Type : MCQ Option Shuffling : No

A practical voltage source is usually represented by _____.

Options :

1. A resistance in series with an ideal current source
2. A resistance in series with an ideal voltage source
3. A resistance in parallel with an ideal current source
4. A resistance in parallel with an ideal voltage source

Under the condition of maximum power transfer from an a.c.source to a variable load, choose the correct option from following, _____

Options :

1. The load impedance must be inductive if the source impedance is inductive.
2. The sum of source impedance and load impedance is zero.
3. The sum of source reactance and load reactance is zero.
4. The load impedance and source impedance has same phase angle and polarity.

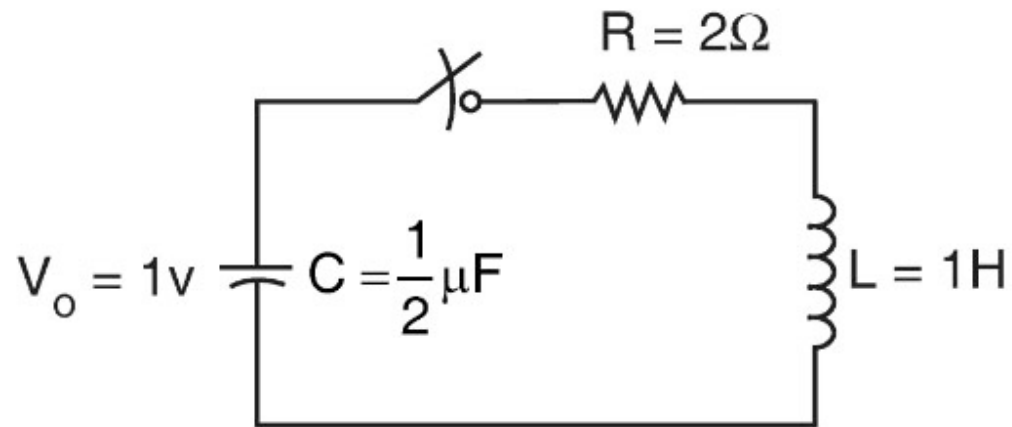
Question Number : 13 Question Id : 150561415 Question Type : MCQ Option Shuffling : No

An R-L-C series circuit consists of a resistance of $1000\ \Omega$, an inductance of 100 mH and a capacitance of 10pF , the value of resonance frequency is _____.

Options :

1. 25 kHz
2. 52 kHz
3. 107 kHz
4. 159 kHz

For a series circuit shown below what will be the correct expression of current $i(t)$?



Options :

1. $i(t) = e^{-t} \sin t \cdot u(t)$
2. $i(t) = e^{-t} \cos t \cdot u(t)$
3. $i(t) = \sin \omega t \cdot u(t)$
4. $i(t) = e^{-t} (\omega t + \theta)$

Question Number : 15 Question Id : 150561417 Question Type : MCQ Option Shuffling : No

Three identical coils connected in delta to a 440v, 3 phase supply take a power of 50 kW and line current of 90A. What will be the power factor of the coils ?

Options :

1. 0.8 lagging
2. 0.8 leading
3. 0.73 lagging
4. Unity

Question Number : 16 Question Id : 150561418 Question Type : MCQ Option Shuffling : No

The unit of permittivity is _____.

Options :

1. Volt/(Ampere-sec)
2. (Coulomb)²/(Newton – m²)
3. Newton/(Coulomb)²
4. Farad/volt

Question Number : 17 Question Id : 150561419 Question Type : MCQ Option Shuffling : No

The electric field intensity due to infinite line charge at any point is _____.

Options :

1. Inversely proportional to square of perpendicular distance from line charge.
2. Proportional to square of perpendicular distance from line charge.
3. Inversely proportional to the perpendicular distance from line charge.
4. Proportional to the perpendicular distance from line charge.

Question Number : 18 Question Id : 150561420 Question Type : MCQ Option Shuffling : No

The Gauss's law in point form for electrostatic field is _____.

Options :

1. $\nabla \times \mathbf{D} = \mathbf{Q}$

2. $\nabla \times \mathbf{D} = \mathbf{E}$

3. $\nabla \cdot \mathbf{D} = \rho_v$

4. $\nabla \cdot \mathbf{D} = 0$

Which of the following equation is correct representation of relationship between voltage (potential) and electric field intensity ?

Options :

1. $\bar{E} = \oint \bar{v} \cdot d\bar{s}$

2. $\bar{E} = \oint \bar{e}_v \cdot d\bar{v}$

3. $v_{AB} = - \int_B^A \bar{E} \cdot d\bar{l}$

4. $\nabla \times \bar{v} = \bar{E}$

Question Number : 20 Question Id : 150561422 Question Type : MCQ Option Shuffling : No

For steady magnetic field $\nabla \times \mathbf{H} = ?$

Options :

1. $\mathbf{B}$ – flux density
2. $\mathbf{D}$ – electric flux density
3. zero
4. $\mathbf{J}$ – electric current density

The inductance per meter length of a co-axial cable of inner radius 'a' and outer radius 'b' is _____.

Options :

1.
$$L = \frac{\mu_o}{2\pi} \ln(b/a) \text{ H/m}$$

2.
$$L = \frac{2\pi}{\mu_o} \ln(b/a) \text{ H/m}$$

3.
$$L = 2\pi \ln\left(\frac{1}{a} - \frac{1}{b}\right) \text{ H/m}$$

4.
$$L = \frac{\mu_o}{2\pi} (\ln(a/b)) \text{ H/m}$$

The electrostatic energy density is given by _____.

Options :

1. $\frac{1}{2} \epsilon E^2$

2. $\frac{1}{2} \epsilon V^2$

3. $\frac{1}{2} \epsilon Q^2$

4. $\frac{1}{2} \epsilon D^2$

The capacitance between two concentric spherical conducting shells of radius 'a' and 'b' ($b > a$) is _____.

Options :

1.
$$C = \epsilon \frac{a}{b} \left(\frac{a^2 - ab}{b^2 - ab} \right)$$

2.
$$C = \frac{2\pi a^2}{b^2} \left(\frac{1}{a} - \frac{1}{b} \right)$$

3.
$$C = \frac{4\pi\epsilon}{\frac{1}{a} - \frac{1}{b}}$$

4.
$$C = \frac{4\epsilon}{a - b}$$

The energy density of steady magnetic field is given by _____.

Options :

1. $\frac{1}{2}B^2$

2. $\frac{1}{2}\mu B^2$

3. $\frac{1}{2}\mu H^2$

4. $\frac{1}{2}H^2$

The reluctance of linear isotropic homogeneous magnetic material of length 'd' and uniform cross section 's' is given by _____.

Options :

1. $R = \frac{\mu s}{d}$

2. $R = \frac{d}{\mu s}$

3. $R = \frac{\mu d}{s}$

4. $R = \frac{s}{\mu d}$

_____ are functions that are completely specified in time. The nature and amplitude of such signal at any time can be predicted.

Options :

1. Deterministic signals
2. Non-deterministic signals
3. Random signal
4. None of the above

Question Number : 27 Question Id : 150561429 Question Type : MCQ Option Shuffling : No

Following is the impulse response of a discrete-time LTI (Linear Time-Invariant) system.

$$h[n] = (5)^n u[3 - n]$$

Which of the following statement is true for this LTI system ?

Options :

1. The system is causal
2. The system is stable
3. The system is causal but unstable
4. The system is not causal and stable

Question Number : 28 Question Id : 150561430 Question Type : MCQ Option Shuffling : No

The signal $x(t) = \cos t + \sin \sqrt{2} t$ is _____.

Options :

1. Periodic
2. Non-periodic
3. Both (1) and (2)
4. None of the above

Question Number : 29 Question Id : 150561431 Question Type : MCQ Option Shuffling : No

The even and odd components of the signal $x(t) = \cos t + \sin t$ are _____.

Options :

1. $\cos t$ and $\sin t$
2. $\sin t$ and $\cos t$
3. $\cos t$ and $\tan t$
4. $\tan t$ and $\cos t$

Question Number : 30 Question Id : 150561432 Question Type : MCQ Option Shuffling : No

The signal $x(t) = A \cos(\omega_0 t + d)$ is a _____.

Options :

1. Power signal
2. Energy signal
3. Both (1) and (2)
4. None of above

Question Number : 31 Question Id : 150561433 Question Type : MCQ Option Shuffling : No

The convolution property of the Continuous-Time Fourier Transform (CTFT) states that convolution in the time domain is equivalent to what operation in the frequency domain ?

Options :

1. Convolution
2. Differentiation
3. Multiplication
4. Addition

Question Number : 32 Question Id : 150561434 Question Type : MCQ Option Shuffling : No

The analog signal is given by $x(t) = 5 \cos 2000\pi t + 3 \sin 6000\pi t + 2 \cos 12000\pi t$
the Nyquist rate for this signal is _____.

Options :

1. 12 kHz
2. 24 kHz
3. 36 kHz
4. 6 kHz

Question Number : 33 Question Id : 150561435 Question Type : MCQ Option Shuffling : No

The impulse response of a digital system is given by $h[n] = a^n u(n)$ The given system is stable for which of the following conditions ?

Options :

1. $a = 1$
2. $a > 1$
3. $a < 1$
4. $a \geq 1$

The system described by the differential equation $\frac{dy(t)}{dt} + 2y(t) = x(t)$ is _____.

Options :

1. Non-linear
2. Linear
3. Both (1) and (2)
4. None of the above

Question Number : 35 Question Id : 150561437 Question Type : MCQ Option Shuffling : No

A continuous time signal is said to be _____ if its amplitude is zero for negative time i.e. $x(t) = 0$ for $t < 0$.

Options :

1. Casual
2. Non-casual
3. Energy and power
4. None of the above

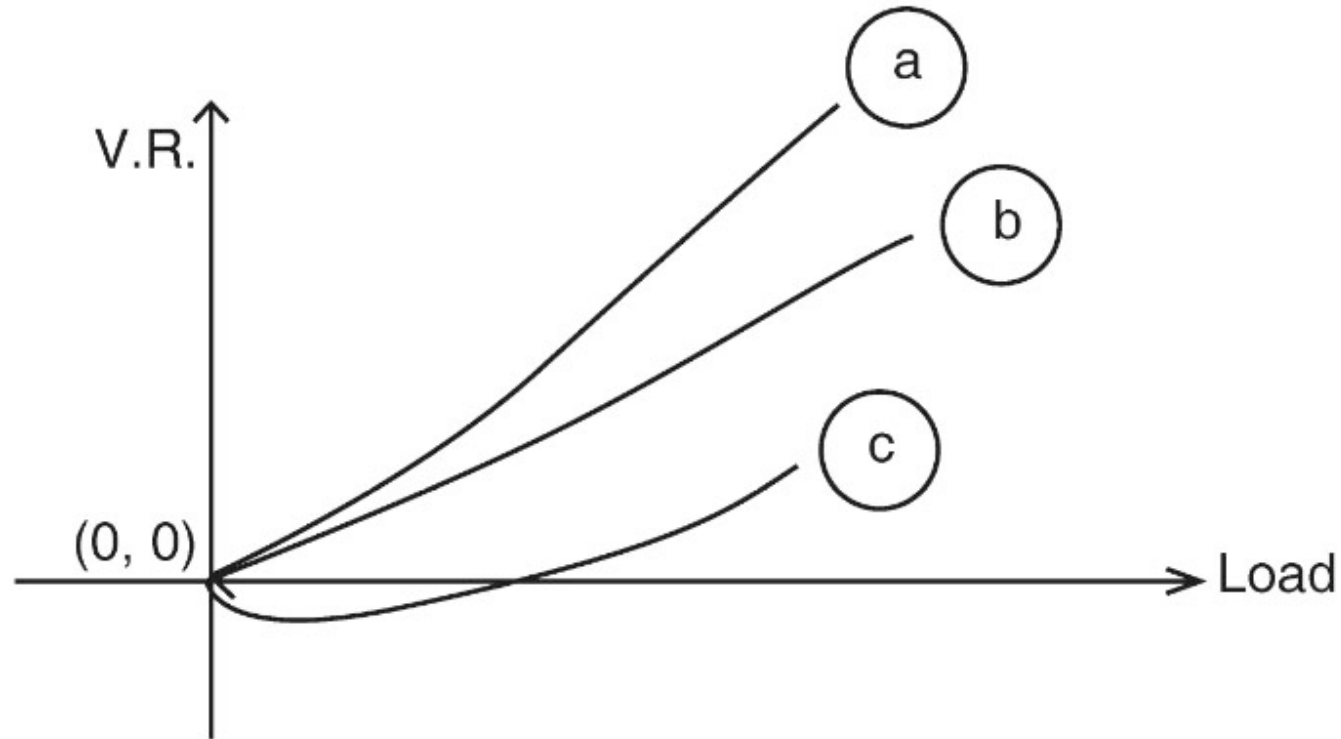
Modern high speed synchronous generator used in thermal power plants have _____.

Options :

1. Salient pole rotors
2. Cylindrical rotor
3. Both (1) and (2)
4. Spherical rotor

Question Number : 37 Question Id : 150561439 Question Type : MCQ Option Shuffling : No

Voltage regulation of an alternator is plotted below :
Identify the nature of load in each case.



Options :

1. (a) Resistive (b) Inductive (c) Capacitive
2. (a) Inductive (b) Resistive (c) Capacitive

3. (a) Capacitive (b) Resistive (c) Inductive

4. (a) Resistive (b) Inductive (c) Load with high frequency harmonics

When synchronous motor is made to operate at variable excitation which of the statements are correct ?

- a. Since applied voltage and load is constant so active component of armature current is constant.
- b. As speed is constant change in dc excitation causes change in excitation voltage E .
- c. Reduction in dc excitation results in lagging power factor.

Options :

- 1. a and b
- 2. b and c
- 3. a and c
- 4. a, b and c

Hunting in synchronous machine can be described as _____ and it can be minimized by _____

Options :

1. Mechanical oscillations in rotor speed about its steady-state speed; use of damper winding
2. Temporal mechanical oscillations in rotor speed; use of external resistance
3. Non-self starting nature of synchronous machine, use of damper winding
4. Non-self starting nature of synchronous machine; use of external resistance

Question Number : 40 Question Id : 150561442 Question Type : MCQ Option Shuffling : No

In three phase induction motor electrical equivalent of mechanical, load in terms of rotor resistance (R_2) and slip (S) is _____.

Options :

1. $\frac{R_2}{S} (1 - S)$

2. $R_2 - \frac{S}{R_2}$

3. $\frac{1}{S(R_2 - 1)}$

4. $R_2(1 + S)$

Question Number : 41 Question Id : 150561443 Question Type : MCQ Option Shuffling : No

A 3 phase, 4 pole induction motor is supplied from 3 phase 415 V, 50 Hz AC source. At 5% slip, the rotor speed is _____.

Options :

1. 3150 rpm
2. 2850 rpm
3. 1575 rpm
4. 1425 rpm

Star Delta starter is used _____.

Options :

1. For 3 phase induction motor having star connected stator and delta connected rotor
2. To connect the rotor of 3 phase induction motor in star at the beginning and later in delta
3. To reduce starting current drawn by 3-phase induction motor
4. All of the above

Question Number : 43 Question Id : 150561445 Question Type : MCQ Option Shuffling : No

The power input to the rotor of a 440 V, 50 Hz, 3 phase 6 pole induction motor is 60 kW. It is observed that the rotor emf makes 60 complete cycles per minute. What will be rotor copper loss ?

Options :

1. 2.5 kW
2. 2.3 kW
3. 1.8 kW
4. 1.5 kW

Question Number : 44 Question Id : 150561446 Question Type : MCQ Option Shuffling : No

In a split-phase single phase induction motor, the main winding and auxiliary winding resistance, respectively are _____.

Options :

1. Low and high
2. Equal
3. High and low
4. Both having zero value

If X_s is the secondary winding reactance of a transformer and 'K' is the ratio of secondary to primary turns of a transformer, the secondary reactance referred to primary side given by _____.

Options :

1. $K.X_s^2$

2. $K^2.X_s$

3. $\frac{X_s}{K}$

4. $\frac{X_s}{K^2}$

Question Number : 46 Question Id : 150561448 Question Type : MCQ Option Shuffling : No

For a linear magnetic system, which of the following is valid ?

Options :

1. Field energy is equal to Co-energy
2. Field energy is greater than Co-energy
3. Field energy is less than Co-energy
4. Co-energy is zero

Question Number : 47 Question Id : 150561449 Question Type : MCQ Option Shuffling : No

Select true statement about a single phase auto transformer.

Options :

1. It contains three windings.
2. It offers the best cost saving for a transformation ratio of 1.
3. It can operate only in step down mode.
4. It automatically corrects and maintains the secondary voltage constant.

Question Number : 48 Question Id : 150561450 Question Type : MCQ Option Shuffling : No

A 230 V dc series motor is taking 50 A. Resistance of armature and series field winding are 0.2 ohm and 0.1 ohm respectively. The mechanical power developed is _____.

Options :

1. 10.75 kW
2. 12.5 kW
3. 7.5 kW
4. 11.5 kW

Question Number : 49 Question Id : 150561451 Question Type : MCQ Option Shuffling : No

A dc shunt generator has an induced emf of 254 V. When machine is loaded, the terminal voltage dropped down to 240 V. Neglecting armature reaction, what will be the load current if armature resistance is 0.04 ohm and field circuit resistance of 240 ohm ?

Options :

1. 350 A
2. 340 A
3. 370 A
4. 320 A

Question Number : 50 Question Id : 150561452 Question Type : MCQ Option Shuffling : No

Following are the advantages of Brushless DC (BLDC) motor except _____.

Options :

1. Long life
2. No sparking
3. Less maintenance
4. Need sensing circuit

Question Number : 51 Question Id : 150561453 Question Type : MCQ Option Shuffling : No

Which of the following is not an assumption made in solving swing equation ?

Options :

1. The resistances of lines and machines are neglected
2. The effect of damper winding is neglected
3. The effect of saliency is neglected
4. All series reactances are neglected

The per-unit method is particularly convenient in power system, because _____.

Options :

1. Various sections of power system are connected through transformers and have same voltage levels.
2. Various sections of power system are not connected through transformers.
3. Various sections of power system are connected through transformers and have different voltage levels.
4. None of the above

A loss less line terminated with its surge impedance has _____.

Options :

1. flat voltage profile.
2. transmission line angle is greater than actual length of the line.
3. transmission line angle is less than actual length of the line.
4. both (1) and (2)

Question Number : 54 Question Id : 150561456 Question Type : MCQ Option Shuffling : No

The condition for economic loading of two generators having costs C_1 and C_2 delivering loads P_1 and P_2 will be _____.

Options :

1. $C_1 = C_2$

2. $\frac{dP_1}{dC_1} = \frac{dP_2}{dC_2}$

3. $P_1 = P_2$

4. $\frac{dC_1}{dP_1} = \frac{dC_2}{dP_2}$

Question Number : 55 Question Id : 150561457 Question Type : MCQ Option Shuffling : No

A plant has maximum demand of 100 MW and installed capacity of station is 125 MW. If station operates at load factor of 0.59. At what capacity factor and utilization factor plant is working ?

Options :

1. 0.59 and 0.85
2. 0.47 and 0.8
3. 0.8 and 0.37
4. 0.85 and 0.59

In case of fault involving ground, which of the following is correct ?

Options :

1. Positive sequence current is not present.
2. Negative sequence current is not present.
3. Zero sequence current is not present.
4. The positive and negative sequence currents balance between themselves and zero sequence current flows through the ground and overhead ground wires.

Question Number : 57 Question Id : 150561459 Question Type : MCQ Option Shuffling : No

A 50Hz, 11 KV generator is connected to a power system. The system reactance and capacitance per phase are 10 mH and 0.02 μ F. Calculate maximum restriking voltage.

Options :

1. 8.98 KV
2. 15.55 KV
3. 7.78 KV
4. 17.96 KV

Question Number : 58 Question Id : 150561460 Question Type : MCQ Option Shuffling : No

In case of short circuit of a loaded synchronous machine if X_d'' , X_d' and X_d is the subtransient, transient and synchronous reactance of the machine then _____.

Options :

1. $X_d'' = \frac{X_d' + X_d}{2}$
2. $X_d'' = X_d' = X_d$
3. $X_d'' < X_d' < X_d$
4. $X_d'' > X_d' > X_d$

Question Number : 59 Question Id : 150561461 Question Type : MCQ Option Shuffling : No

The current chopping tendency in SF_6 circuit breaker is minimised by using SF_6 gas at relatively _____.

Options :

1. High pressure, low velocity
2. High pressure, high velocity
3. Low pressure, high velocity
4. Low pressure, low velocity

In Gauss-Seidel load flow method, a PV bus is treated as PQ bus _____.

Options :

1. When the voltage magnitude at that bus is not equal to 1 pu.
2. When the reactive power at that bus violates the lower or upper limit.
3. PV bus is never treated as PQ bus.
4. PV bus is always treated as PQ bus.

In case of three-stepped distance protection of transmission lines, which of the following is correct ?

Options :

1. Zone – 3 is used as primary protection for 80% of the protected feeder.
2. Zone – 3 is used as primary protection for 20% of the protected feeder.
3. Zone – 3 is used as back-up protection only.
4. None of the above

Consider following statements :

Statement X : Series compensation improves both steady state stability as well as transient stability.

Statement Y : Series compensation is effective only during heavy loads. During off-peak conditions, shunt compensation has to be provided.

Which of the following is correct ?

Options :

1. Statement X is true but Statement Y is false.
2. Statement X is false but Statement Y is true.
3. Both Statement X and Statement Y are true.
4. Both Statement X and Statement Y are false.

Which of the statements are correct for synchronous condenser when used for reactive power support ?

- a. Synchronous condenser can supply KVAR equal to rating and absorb KVAR upto 50% of its capacity.
- b. It adds up short circuit capacity of system and increases breaker rating.
- c. Momentary drop in voltage causes synchronous condenser to supply loss KVARs.

Options :

- 1. a, b, c are correct
- 2. a and b are correct
- 3. b and c are correct
- 4. a and c are correct

While formulation of Z_{bus} upon adding new branch; type Z modification has resulted due to _____.

Options :

1. Z_b is added from new bus to reference bus
2. Z_b is added from a new bus to an old bus
3. Z_b connects an old bus to reference bus
4. Z_b connects two old buses

Question Number : 65 Question Id : 150561467 Question Type : MCQ Option Shuffling : No

Which of the following is not a method of representing load ?

Options :

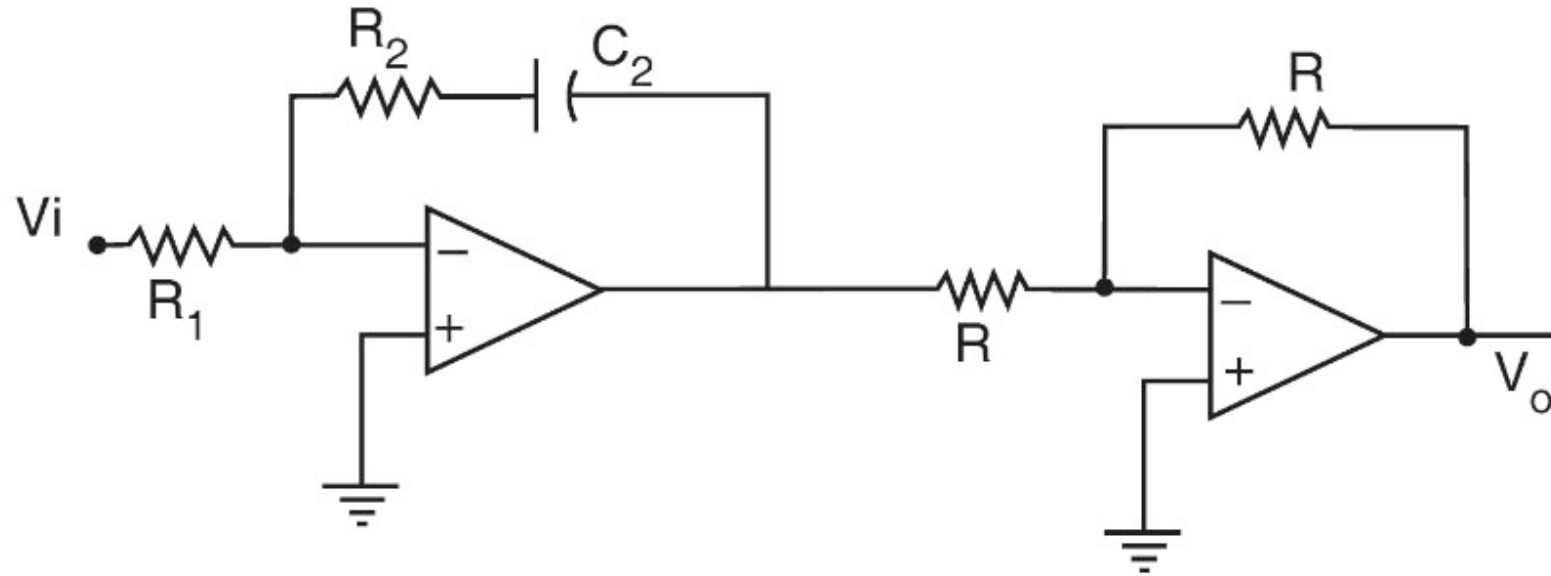
1. Constant impedance representation
2. Constant current representation
3. Constant voltage representation
4. Constant power representation

The addition of zero to the open-loop transfer function has the effect of _____.

Options :

1. pulling the root locus to the left tending to make the system more stable.
2. to speed up the settling of response.
3. both (1) and (2) are correct
4. none of the above correct

An Op-Amp circuit shown below can represent a controller as mentioned in options.



Options :

1. Proportional Controller
2. Proportional Plus Derivative (PD) Controller
3. Proportional Integral (PI) Controller
4. Proportional Integral Derivative (PID) Controller

A few important properties of signal flow graphs are as follows :

1. A branch indicates the functional dependence of one signal on another.
2. A node adds the signals of all incoming branches and transmit this sum to all outgoing branches.
3. For a given system, a signal flow graph is not unique.

Options :

1. 1, 2 and 3 are correct
2. 1 is correct and 2 and 3 are false
3. 1 and 2 are correct and 3 is false
4. 1, 2 and 3 are false

The transfer function of a controller is given as,

$$G_c(s) = K_p \left[1 + \frac{1}{T_1 s} + T_2 s \right]$$

Where T_1 and T_2 are the time constants. The controller represents _____.

Options :

1. PI controller
2. PID controller
3. PD controller
4. P controller

A system is described by the transfer function, $T(s) = \frac{200}{s^4 + 6s^3 + 11s^2 + 6s + 200}$

Use R-H criterion to find number of poles on left half (LH) plane, right half (RH) plane and on jw axis.

Options :

1. Two poles on RH plane, No poles on LH plane, no poles on jw axis
2. Two poles on RH plane, two poles on LH plane, no poles on jw axis
3. No poles on RH plane, two poles on LH plane, two poles on jw axis
4. Two poles on RH plane, two poles on LH plane, two poles on jw axis

If improvement in both transient response and steady-state response are desired, then _____.

Options :

1. lead compensator is used.
2. lag compensator is used.
3. both a lead and lag compensator are used simultaneously.
4. none of the above correct

A second order system is having a transfer function, $G(s) = \frac{64}{s^2 + 8s + 64}$

The damping ratio and natural frequency will be _____.

Options :

1. 8 and 1 rad/s
2. 1 and 8 rad/s
3. 0.5 and 8 rad/s
4. 8 and 0.5 rad/s

For a first order system transfer function is given by, $G(s) = \frac{20}{s+20}$

When a unit step input is applied, the Time constant (T), Settling time (T_s) and Rise time (T_r) will be _____.

Options :

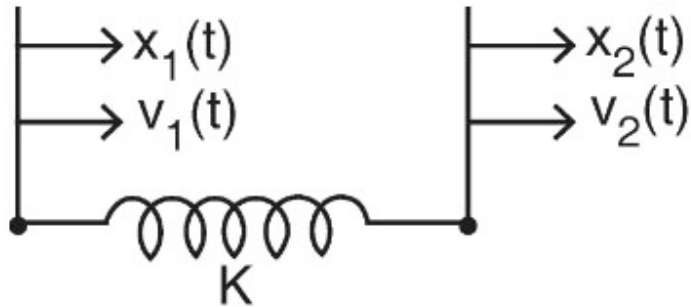
1. $T = 0.2, T_s = 0.11 \text{ sec}, T_r = 0.05 \text{ sec}$

2. $T = 0.11, T_s = 0.2 \text{ sec}, T_r = 0.05 \text{ sec}$

3. $T = 0.2, T_s = 0.05 \text{ sec}, T_r = 0.11 \text{ sec}$

4. $T = 0.05, T_s = 0.2 \text{ sec}, T_r = 0.11 \text{ sec}$

For the spring element, shown below, which of the relations is correct ?



Options :

1. $F = K (v_1 - v_2) = K (\dot{x}_1 - \dot{x}_2)$

2. $F = K \int_{-\infty}^t (v_1 - v_2) dt = K (x_1 - x_2)$

3. $F = K \frac{d(v_1 - v_2)}{dt} = K \frac{d^2(x_1 - x_2)}{dt^2}$

4. $F = K \int_{-\infty}^t (v_1 - v_2) dt = K (\dot{x}_1 - \dot{x}_2)$

Question Number : 75 Question Id : 150561477 Question Type : MCQ Option Shuffling : No

For a system, the steady state error for step input, ramp input and parabolic input is 0, $\frac{R}{K}$ and ∞ respectively, where K is constant. From the steady state errors it seems that the given system must be _____.

Options :

1. Type 0
2. Type 1
3. Type 2
4. None of the above

_____ bridge is used for measuring an unknown inductance in terms of a known capacitance and resistance.

Options :

1. Maxwell's L/C
2. Hay's
3. Owen
4. Anderson

Question Number : 77 Question Id : 150561479 Question Type : MCQ Option Shuffling : No

Which statement is true about thermal imaging camera ?

Options :

1. It is used in surveillance, temperature measurement.
2. It takes images even in dark, smoke, mist.
3. It detects radiance of a surface which is a function of its temperature.
4. All of these

Question Number : 78 Question Id : 150561480 Question Type : MCQ Option Shuffling : No

What are the functions and capabilities of a power quality analyzer ?

Options :

1. Measurement and analysis
2. Trouble shooting, monitoring and logging
3. System compatibility and remote control
4. All of the above

It is necessary to correct the difference between k and k_n . It can be done with the help of ratio error, which is defined as : (k and k_n are the actual ratio and nominal ratio of an instrument transformer) _____.

Options :

1.
$$\epsilon = \frac{k_n - k}{k}$$

2.
$$\epsilon = \frac{k}{k_n - k}$$

3.
$$\epsilon = \frac{k - k_n}{k}$$

4.
$$\epsilon = \frac{k}{k - k_n}$$

Match the analog electrical instrument with measurand.

A Only DC current

A' Dynamometer type moving coil type

B AC voltage

B' Permanent magnet moving coil type

C AC power

C' Induction type

D AC energy

D' Moving iron type

Options :

1. AA' BC' CB' DD'

2. AB' BD' CA' DC'

3. AB' BC' CD' DA'

4. AA' BB' CC' DD'

Question Number : 81 Question Id : 150561483 Question Type : MCQ Option Shuffling : No

In voltage follower circuit what is the operational sequence ?

Options :

1. Output is feedback to inverting terminal, input is given to non-inverting terminal.
2. Output is feedback to non-inverting terminal input is given to inverting terminal.
3. Inverting terminal is grounded and input is given to non-inverting terminal.
4. Inverting terminal is supplied and not inverting terminal grounded.

If OPAMP is used in typical subtractor mode with R_1 is supply side resistor and R_2 feedback resistor. The output voltage is given by _____.

Options :

1. $(V_1 - V_2)$

2. $\left(\frac{R_2}{R_1}\right) (V_1 - V_2)$

3. $\left(\frac{R_2}{R_1}\right) (V_2 - V_1)$

4. $\left(\frac{R_1}{R_2}\right) (V_2 - V_1)$

Question Number : 83 Question Id : 150561485 Question Type : MCQ Option Shuffling : No

A timer circuit is constructed through 555. This circuit is developed for astable operation. For $V_{cc} = 12\text{ V}$, $R_1 = 10\text{ kilo ohm}$, $R_2 = 2.3\text{ kilo ohm}$ and $C = 0.1\text{ micro farad}$. The output frequency for this is 1977 Hz . What will be duty cycle ?

Options :

1. 50%
2. 84%
3. 78%
4. 65%

Question Number : 84 Question Id : 150561486 Question Type : MCQ Option Shuffling : No

Maximum efficiency for a series fed Class A amplifier can be _____.

Options :

1. 40%
2. 25%
3. 16%
4. 30%

Wein bridge oscillator is designed such a way that all branch resistors are chosen as 'R' ohms and capacitors are chosen as 'C' micro farad. Then resulting oscillation frequency is given by _____.

Options :

1.
$$f_0 = \frac{1}{2\pi\sqrt{RC}}$$

2.
$$f_0 = \frac{1}{2\pi\sqrt{R/C}}$$

3.
$$f_0 = \sqrt{R/C}$$

4.
$$f_0 = \frac{1}{2\pi(RC)}$$

SCRs with rating of 1200 V and 250 A are used in a string to handle 5KV and 2KA, calculate the number of series and parallel units required in case of derating factor 0.2.

Options :

1. $n_s = 6$ and $n_p = 10$

2. $n_s = 5$ and $n_p = 9$

3. $n_s = 7$ and $n_p = 11$

4. $n_s = 4$ and $n_p = 8$

Question Number : 87 Question Id : 150561489 Question Type : MCQ Option Shuffling : No

For a single phase half bridge inverter feeding a resistive load of $R = 7.5 \Omega$.

The d.c. voltage of the inverter is given as 220 V. Determine RMS value of the fundamental component of output voltage.

Options :

1. 89.08
2. 99.08
3. 79.08
4. 69.08

In a PWM inverter, feedback diodes are required to _____.

Options :

1. transfer the energy stored in the load inductance back to the dc source.
2. provide a close-loop feedback control.
3. provide an alternate path for current when the switch has damaged.
4. none of the above

In a voltage source inverter, a diode is connected in anti parallel with the controlled IGBT switch to _____.

Options :

1. prevent reversal of d.c. link current.
2. protect the circuit against accidental reversal of d.c. bus polarity.
3. allow a non-unity power factor load at the output.
4. none of these

Question Number : 90 Question Id : 150561492 Question Type : MCQ Option Shuffling : No

In a current source inverter _____.

Options :

1. Free wheeling diode must be used.
2. Diode in series with the switching transistor must be used.
3. Feedback diodes must be used.
4. All of the above are a must.

To drive a mechanical load at constant speed in steady-state condition, motor torque must be equal to _____.

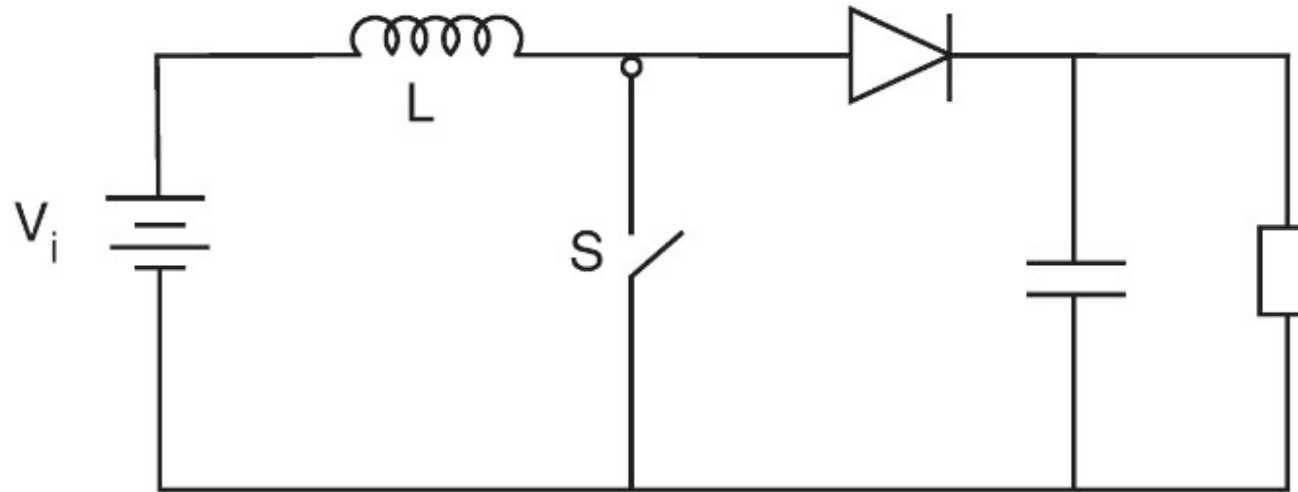
- a. Windage torque + load torque
- b. Frictional torque + windage torque + load torque
- c. Load torque + frictional torque
- d. None of the above

Options :

- 1. d
- 2. b
- 3. a
- 4. c

Question Number : 92 Question Id : 150561494 Question Type : MCQ Option Shuffling : No

Find the output voltage of the following converter, if duty factor of the switch S is 0.4. Input voltage $V_i = 100$ V.



Options :

1. 60 V
2. 166.66 V
3. 250 V
4. None of these

Question Number : 93 Question Id : 150561495 Question Type : MCQ Option Shuffling : No

In boost converter output voltage will be decided by _____.

Options :

1. Value of series inductor and input voltage.
2. Switching frequency of chopper and value of inductor.
3. Value of inductor and rate of change of source current.
4. Value of inductor and load current.

In a PWM inverter fed induction motor drive, the motor frequency can be adjusted by controlling the _____.

Options :

1. Frequency of carrier wave of PWM.
2. Frequency of reference sine wave of PWM.
3. Hysteresis band width.
4. None of the above

Match the following :

- | | |
|-----------|---|
| A. BJT | a. Faster switching |
| B. MOSFET | b. HV, high current application |
| C. IGBT | c. Limited to low frequency application |

Options :

1. A – b B – c C – a
2. A – a B – b C – c
3. A – c B – a C – b
4. None of the above

Vertically oriented structured non-punch IGBT has which of the following layers present ?

- a. Source layer
- b. Body region layer
- c. Drift layer
- d. Buffer layer
- e. Injection layer

Options :

1. a, c, d, e
2. a, b, c, d, e
3. a, b, c, d

4. a, b, c, e

Question Number : 97 Question Id : 150561499 Question Type : MCQ Option Shuffling : No

An IGBT has _____.

Options :

1. High input impedance and low on-state conduction loss.
2. Low input impedance and high on-state conduction loss.
3. Both input impedance and on-state conduction loss are high.
4. Both input impedance and on-state conduction loss are low.

Question Number : 98 Question Id : 150561500 Question Type : MCQ Option Shuffling : No

A dual converter is operated in circulating current mode. Set 1 operates with firing angle α . Then, firing angle of set 2 is _____.

Options :

Set 2 is not fired

1.

2. α

3. $\frac{\pi}{2} + \alpha$

4. $\pi - \alpha$

A single phase fully controlled converter is supplying R – L load. A free wheel diode is connected across the load. The average load voltage is given by _____.

Options :

1.
$$\frac{\sqrt{2}V_m}{\pi} \cos \alpha$$

2.
$$\frac{2\sqrt{2}V_{rms}}{\pi} (1 + \cos \alpha)$$

3.
$$\frac{2\sqrt{2}V_{rms}}{\pi} \cos \alpha$$

4.
$$\frac{V_m}{2\pi} (1 + \cos \alpha)$$

For a three phase fully controlled converter supplying resistive load. For any value of firing angle for 50 Hz supply. The output voltage contains ripple frequency of _____.

Options :

1. 300 Hz
2. 100 Hz
3. 50 Hz
4. Zero Hz